

Correction des anciens examens de Graph

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1 Examen June 2024

1.1 Q1.1 : Definition of selection and social influence

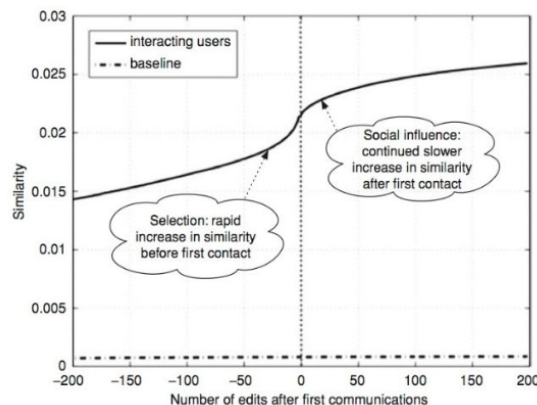
- **Selection**: People select friends with similar characteristics (an internal mechanism).
- **Social influence**: People modify their behaviors to be closer to their friends (an external mechanism), also called "peer pressure".

1.2 Q1.2 : Wikipedia as a social network

- **Wikipedia** is a collaborative encyclopedia written by editors.
- We define the **social network** as follows:
 - Two editors are linked if one has communicated on the other's talk page.
 - An editor's **behavior** consists of the set of articles they have edited.
- We define **similarity between two editors** through their behaviors:
 - $$S(A, B) = \frac{\text{Number of articles edited by both A and B}}{\text{Number of articles edited by at least one A and B}}$$

1.3 Q1.3 : Comparing selection and social influence on Wikipedia

- We show average similarity of two editors relative to the instant when they first communicated.
- Similarity increases faster and faster until editors meet (they are independently becoming more similar, which shows **selection**).
- After they meet, similarity increases slower as the editors adapt to each other (e.g., divide the work between them), which shows **social influence**.
- This curve shows average over many editor; for individual editors, it is much less smooth.

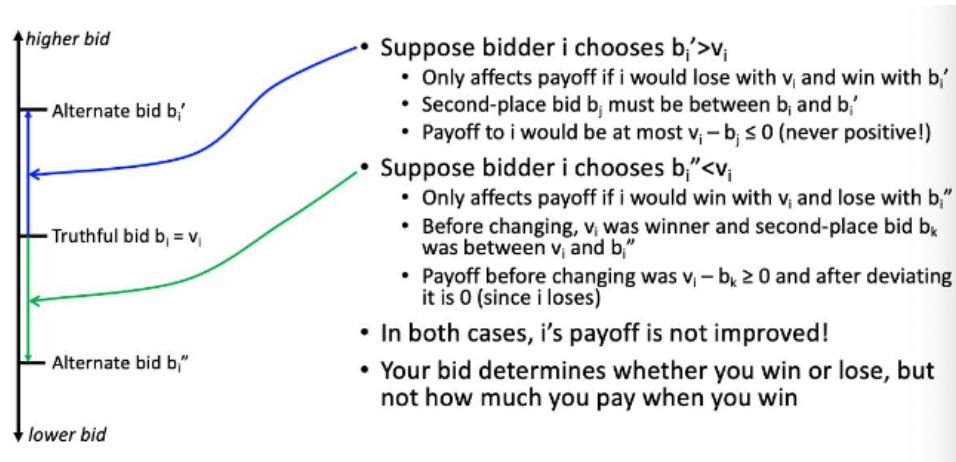


1.4 Q2.1 : Second price auction in the language of Game Theory

- In a **second price auction**, the payoff of each bidder i depends on whether they win or lose:
 - If bidder i does **not** submit the highest bid, their payoff is 0.
 - If bidder i submits the highest bid (i.e., wins), and the **second-highest bid** is b_j , then their payoff is $v_i - b_j$.

1.5 Q2.2 : Second Price auction theorem

- **Second-Price Auction Theorem:** In a sealed-bid second-price auction, it is a **dominant strategy** for each bidder i to choose a bid $b_i = v_i$



- In both cases, bidder i 's payoff is not improved.
- Your bid determines whether you win or lose, but not how much you pay when you win.

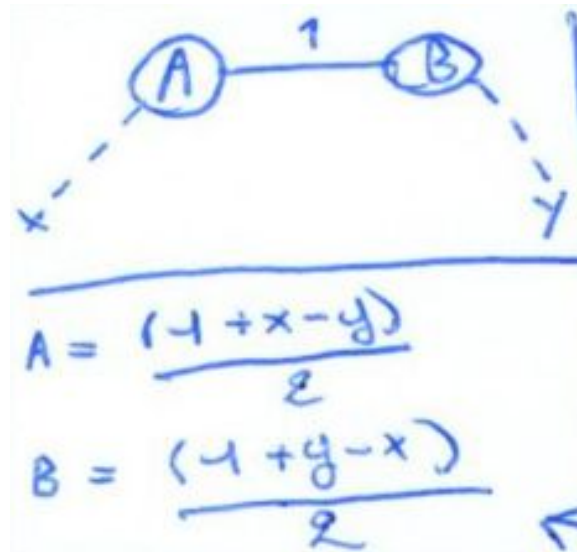
1.6 Q3 : The three-node path

- Node **B** is very powerful because it can exclude **A** or **C** — they are completely dependent on it.



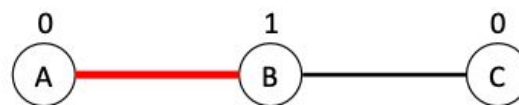
1.7 Q3.2 : The three principles of network negotiation

- **Nash bargaining solution:** Indicates that the division is done in such a way that node **A** gets what is due to it based on its position; this also applies to node **B** according to the distribution shown here.

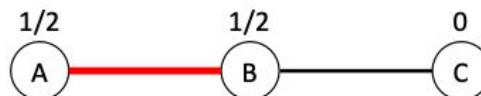


1.8 Q3.3 : Stable outcome

- **Stable outcome:** No node can increase its payoff by changing its trading partner.



Stable, because there is nothing C can offer to B to improve B's value!



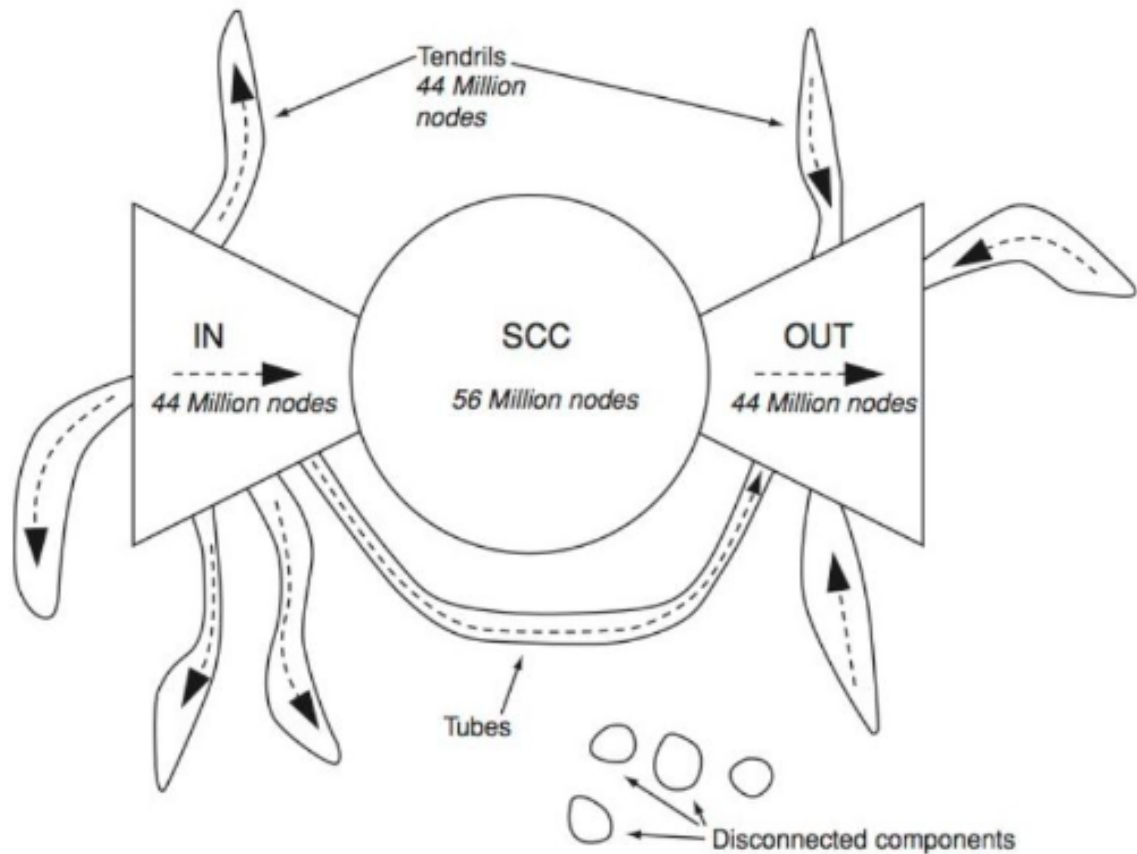
Unstable, because C can offer 2/3 to B and keep 1/3, so both are better off!

1.9 Q3.4 : Balanced outcomes

- **Balanced outcome:** Each resource allocation is a Nash bargaining solution.

Image: (dans feuille)

1.10 Q4 : Explain Bow-Tie Structure of the Web



Disconnected components (17 Million nodes)

1.11 Q5: Bayes' Rule

Let:

- Y : the cab is yellow
- B : the cab is black
- S_Y : the witness says the cab is yellow

We want to find:

$P(Y | S_Y)$ = Probability the cab is yellow given the witness says yellow

Bayes' Rule:

$$P(Y | S_Y) = \frac{P(S_Y | Y) \cdot P(Y)}{P(S_Y)}$$

Given:

$$\begin{aligned} P(Y) &= 0.10, & P(B) &= 0.90 \\ P(S_Y | Y) &= 0.75, & P(S_Y | B) &= 0.25 \end{aligned}$$

Total Probability:

$$\begin{aligned} P(S_Y) &= P(S_Y | Y) \cdot P(Y) + P(S_Y | B) \cdot P(B) \\ P(S_Y) &= (0.75 \cdot 0.10) + (0.25 \cdot 0.90) = 0.075 + 0.225 = 0.30 \end{aligned}$$

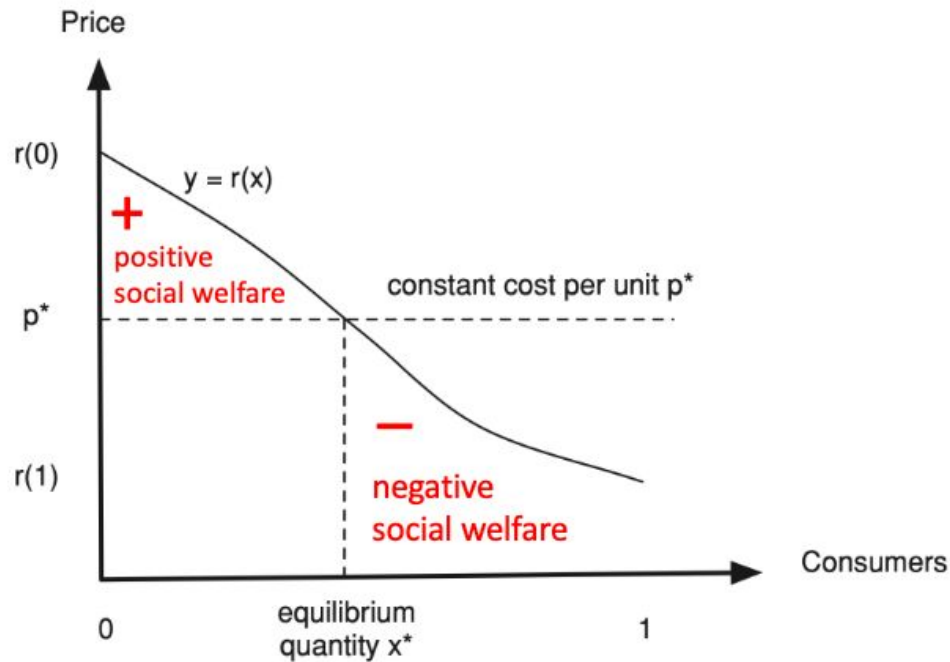
Applying Bayes' Rule:

$$P(Y | S_Y) = \frac{0.75 \cdot 0.10}{0.30} = \frac{0.075}{0.30} = 0.25$$

1.12 Q6: Market Without Direct Benefit Effects

The firm sells at least at P^* , otherwise it **loses money**. It's in **equilibrium** because for every $x < x^*$, we have $r(x) > r(x^*)$, and conversely, if $x^* < x$, then $r(x^*) > r(x)$.

The **pressure on consumers** makes this equilibrium a **stable equilibrium**. It truly is stable because the **area under the curve is maximized**.



1.13 Q6.2: Market With Direct-Benefit Effects

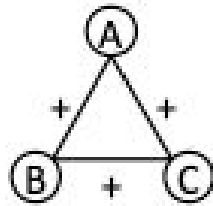
- **Formula:** $r(x) \cdot f(z)$
- **Assume:**
 - $r(x) = 1 - x$ (Decreases from 1 to 0)
 - $f(z) = z$ (Increases from 0 to 1)
- **Image:** (feuille)
- **There are 3 equilibria:** 0, z' , and z''
 - If $z < z'$: $r(z) \cdot f(z) > r(z') \cdot f(z') \Rightarrow$ weak pressure on consumers $\Rightarrow$ 0 is stable
 - If $z' < z < z''$: $r(z') \cdot f(z') > r(z) \cdot f(z) > r(z'') \cdot f(z'') \Rightarrow$ strong pressure on consumers $\Rightarrow z'$ is unstable (This is a critical point, also called a **tipping point**: we want to move past it)
 - If $z'' < z$: $r(z'') \cdot f(z'') > r(z) \cdot f(z) \Rightarrow$ weak pressure on consumers $\Rightarrow z''$ is stable

2 Examen June 2023

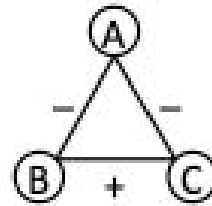
2.1 Q1 : Definition of Structural balance

A **labeled complete graph** is **balanced** if, for each triangle in the graph (i.e., a triple of 3 nodes), the links are :

- $(+++)$ — Three mutual friends
- $(+-)$ — Two friends with a common enemy

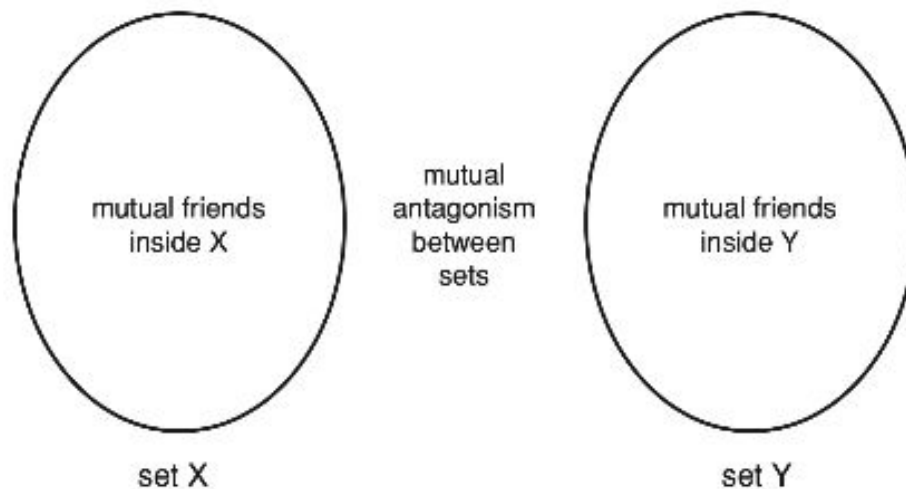


Natural situation
(three mutual friends)



Natural situation
(two friends with common enemy)

2.2 Q1.2 : Structural balance theorem



If a **labeled complete graph** is **balanced**, then either:

- All pairs of nodes are friends,
- or the nodes can be divided into two groups such that:
 - Nodes within the same group are friends
 - Nodes between the two groups are enemies

Proof: Suppose we have a **labeled complete graph** that is **balanced**.

- If the graph has **no negative links at all**, the proof is done.

- Otherwise, pick any node A in the graph.

We define two sets of nodes, X and Y , such that:

- 1. Each set consists of mutual friends, and the sets are enemies.
- 2. Set X is defined as node A and all of its friends.
- 3. Set Y is defined as all enemies of A .

Then:

- Every two nodes in X are friends
- Every two nodes in Y are friends
- Every node in X is an enemy with every node in Y

2.3 Q2.1 : Second-Price auction in the language of Game Theory (Look question Q2.1 Examen June)

2.4 Q2.2 Second-Price auction theorem (Look question Q2.2 Examen June 2024)

2.5 Q3 : Market clearing

In **economics**, **market clearing** means that the **supply of an item is equal to the demand**, so there are **no leftovers of either**.

2.6 Q3.2 : Perfect Matching

If there are an **equal number of nodes on each side of a bipartite graph**, then a **perfect matching** is an assignment between the two sides such that:

- 1. Each node is connected to an edge
- 2. No two left-side nodes are assigned to the same right-side node

2.7 Q3.3 : Preferred-Seller graph

The **buffer payoff** is defined as:

$$V_{ij} - P_i$$

where V_{ij} is the **valuation that buyer j assigns to good i** , and P_i is the price of good i .

The **preferred seller graph** is a **buyer-seller bipartite graph** where: **Each buyer announces its preferred good**, i.e., the one that **maximizes its payoff**.

2.8 Q3.4 : Social optimality

Consider M to be a perfect matching in the preferred seller graph.

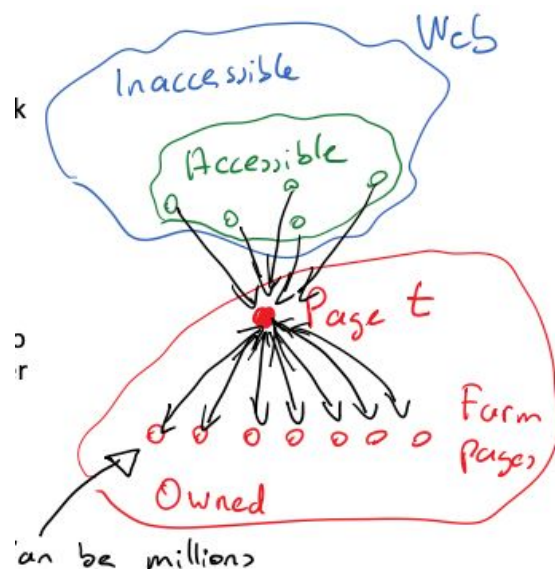
- 1. Total payoff of sellers and buyers = total valuation of buyers
 - (total payoff of sellers) = (sum of all buyers $V_{ij} - P_i$) + (Sum of all seller P_i)
 - Since all "houses" get bought the $-P_i$ and $+P_i$ cancel
 - (total payoff of sellers) = (Sum of V_{ij}) + (Total valuation of buyers)
- 2. Total valuation of buyers is maximized
 - (Total payoff of buyers) = (Sum of all buyers $V_{ij} - P_i$) = (Sum of V_{ij}) - (Sum of P_i)
 - (Total payoff of buyers) = (Total valuation of buyers) - (Sum of all prices)
 - But (Sum of all prices) is a constant
 - Each buyer maximizes its own payoff, so total valuation of buyers is also maximized
- 3. Total payoff of sellers and buyers is maximized

2.9 Q4 : Link farm definition

A **link farm** is a web structure designed to artificially boost the PageRank score of a given webpage t .

It uses farm pages to inflate its score by:

- Establishing **bi-directional links** between t and the farm pages,
- Using **links from unrelated pages** to reinforce the effect.



2.10 Q4.2 : Operation of a link farm

A **link farm** is a collection of interconnected web pages designed to manipulate the PageRank algorithm, which is based on a **random walk model**.

In this model, a "random surfer" either:

- follows a **hyperlink** with a certain probability (typically 85%),
- or **teleports** to a random page with the remaining probability (typically 15%).

Link farms exploit this by creating a **densely connected graph** of pages that continually link to one another. This structure causes **PageRank** to accumulate within the farm.

One or more of these pages then link to a target third-party page, effectively transferring inflated PageRank to that target.

The **teleportation mechanism** does limit the power of link farms to trap all PageRank indefinitely, but as long as the random surfer follows links most of the time, **link farms can still significantly influence rankings**.

While technically feasible, offering link farm services is against Google's Terms of Service $\Rightarrow$ **Illegal**.

2.11 Q5 : Market without direct benefit effects ((Look question Q6 Examen June 2024)

2.12 Q5.2 : Market with direct benefit effects (Look question Q6.2 Examen June 2024)

3 Examen May 2022

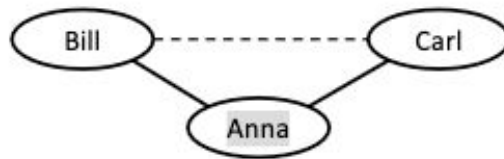
3.1 Q1 : Definition of social-affiliation network

- **Social-affiliation network**: Merges a social network and an affiliation network.
 - Two kinds of nodes: **people** and **foci**.
 - Two kinds of edges: **person-person edge**, **person-focus edge**.

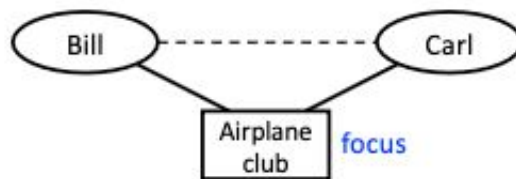
Image feuille

3.2 Q2 : Closure in a social affiliation network

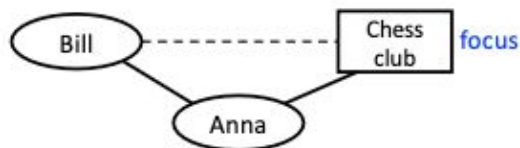
Triadic closure: Bill and Carl become friends because they have a common friend Anna.



Focal closure: Bill and Carl become friends because they have a common focus — the Airplane Club.



Membership closure: Bill joins the Chess Club because his friend Anna is a member. (A friend and a focus have a common friend)



3.3 Q3 : Weak Structural balance

A **labeled complete graph** is **weakly balanced** if for each triangle in the graph (i.e., a triple of 3 nodes), the links are: $(+++)$, $(+-)$, or $(--)$.

- $(+++)$: Three mutual friends — **balanced**.
- $(+-)$: Two friends with a common enemy.
- $(--)$: **less stable in time**, an alliance could be made, but it is **still +- stable** = **weakly balanced**.

3.4 Q3.2 : Weak structural balance theorem

Weak balance theorem: If a labeled complete graph is weakly balanced, then its nodes can be divided into groups such that:

- Any two nodes in the same group are friends
- Any two nodes in different groups are enemies

Proof: We use the same approach as in the proof of the balance theorem.

- Pick any node A of the graph and all its friends (we call this group X).
- Remove X from the graph.
- Look at the set of remaining nodes: it is strictly smaller than the original set.
 - If the set is empty, we are done.
 - Otherwise, continue the proof with the set.
- Ultimately, we arrive at a partition of the original graph into n nodes that satisfies the theorem.

3.5 Q4 : Bargaining in networks

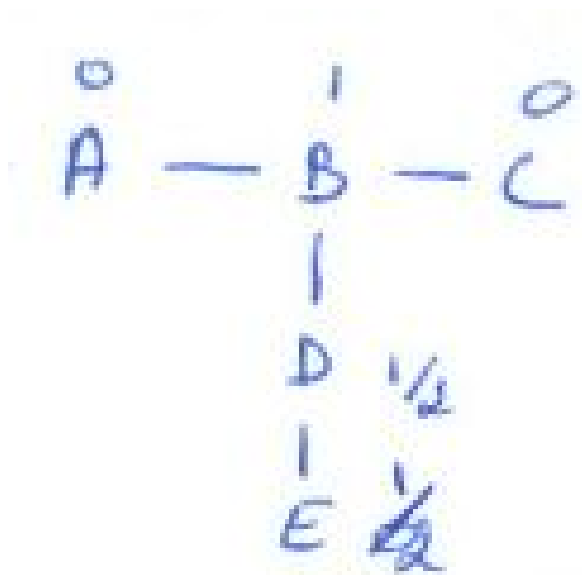
Stable outcome: No node can increase its payoff by changing its trading partner.

Balanced outcome: Each resource allocation is a Nash Bargaining solution.

Nash Bargaining Solution: Indicates that the division is done in such a way that node A gets what's due to it based on its position; this also applies to node B.

$$A = \frac{1 + x - y}{2}$$
$$B = \frac{1 + y - x}{2}$$

3.6 Q5 : Bargaining in a T network



In this network, **node B has a monopoly position** over nodes A and C, which allows it to **leverage both to maximize its gain**.

Node **D is in an intermediate position** between B and E, but since **E has an advantage** thanks to A and C, **D can only negotiate with E**.

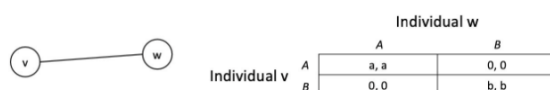
This results in a **symmetric solution** of:

$$\frac{0 + 1 + 0}{2} = \frac{1}{2} \quad (\text{Nash})$$

3.7 Q6 : Link farms definition (Look at question Q4 Examen May 2023)

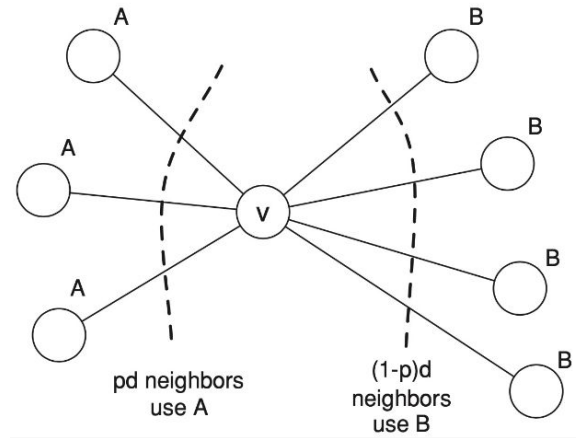
3.8 Q6.2 : Operation of a Link farms definition (Look at question Q4.2 Examen May 2023)

3.9 Q7 : Cascade diffusion model



- We start by looking at **two individuals V and W linked by an edge**.
- Each individual has a choice between **two behaviors, A and B**.
- The **edge gives them an incentive to have matching behaviors**.
- We define a **simple coordination game**:
 - Individuals *v* and *w* are the **players**.
 - Behaviors A and B are the **strategies**.

Looking at all neighbors:

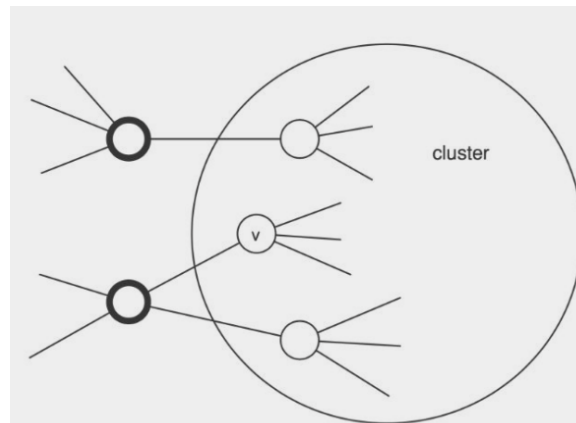


We have the condition:

$$p \geq \frac{b}{a+b}$$

We say that $q = \frac{b}{a+b}$ is the **threshold**.

3.10 Q7.2 : The cluster-cascade theorem



Consider an arbitrary network in which **behavior A** is spreading with **threshold q** , from a set of **initial adopters**.

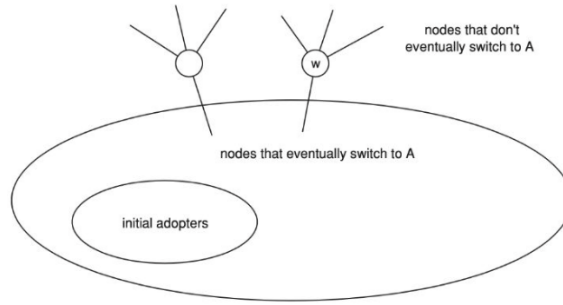
- Suppose the remaining network has a **cluster of density** $> (1 - q)$.

Assume the opposite: v has adopted A at the earliest time step t .

- At that time, the **only neighbors of v using A were outside of the cluster**.

Assume a set of initial adopters fails to cause a **complete cascade**.

- The **cascade stops** because there are still **nodes using B that do not want to switch**.



Let S denote the set of nodes using **behavior B** at the end that do not switch.

- Consider any node w in S .
- Since w does not want to switch, the **fraction of its neighbors using A is less than q** .
- Therefore, the **fraction using B is greater than $(1 - q)$** .
- **Therefore, S is a cluster of density $> (1 - q)$.**

4 Examen Aout 2022

4.1 Q1 : Definition of structural balance (Look question : ...)

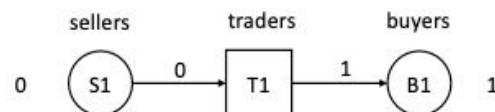
4.2 Q1.2 : Structural balance Theorem (Look question : ...)

4.3 Q2 : Second price auction in the language of Game theory (Look question : ...)

4.4 Q2.2 : Second price auction theorem (Look question ...)

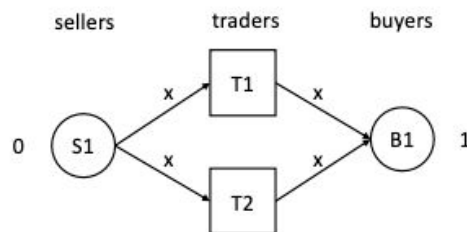
4.5 Q3 : Trading networks with intermediaries

Monopoly



- Buyers and sellers are subject to **monopoly** when they access only a single trader.
- The only **equilibrium** is for the trader to set bid 0 to the seller and ask 1 to the buyer.
- For any other bid and ask between 0 and 1, the trader could **slightly lower the bid or raise the ask**, thereby **achieving higher profit**.

Perfect Competition



- Buyers and sellers are subject to **perfect competition** when they access more than one trader.
- There is **competition** between traders T1 and T2 to buy the copy of the good from S1 and sell it to B1.
- Equilibrium for T2 occurs at **common bid and ask of X**.

4.6 Q4 : Bow-tie structure of the web (Look question ...)

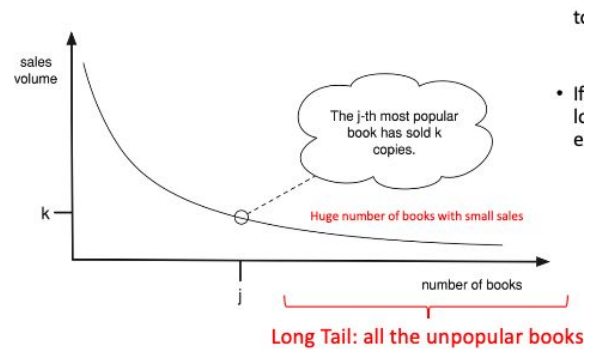
4.7 Q5 : Bayes's rules (Look question ...)

4.8 Q6 : The long Tail

The ability to carry huge inventories is a **big reason for their success**. Huge inventories are much easier to implement on a website than in a brick-and-mortar store.

The Long Tail:

- **Power laws** mean that **popularity decreases very slowly**.
- So, a huge number of unpopular pages can have a large aggregate popularity.
- The success of many websites depends on **having a huge inventory**.



4.9 Q7 : Cascade diffusion model (Look question : ...

4.10 Q7.2 : The cluster-cascade theorem (Look question : ...

5 Examen June 2021

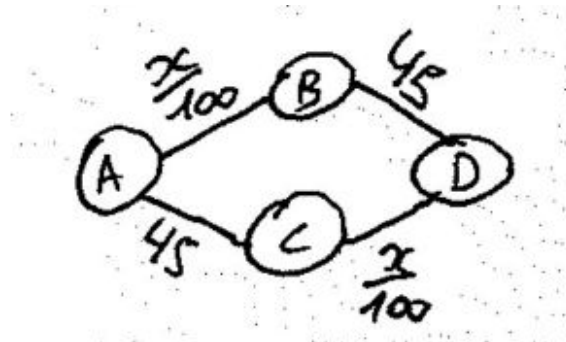
5.1 Q1 : Definition of Braess's Paradox

Braess's paradox refers to the phenomenon where adding a road (or strategy) can worsen the overall situation at equilibrium due to selfish routing. Therefore, adding a resource does not always improve the outcome.

However, adding a road does not necessarily worsen equilibrium performance.

5.2 Q2 : Example of Braess's paradox

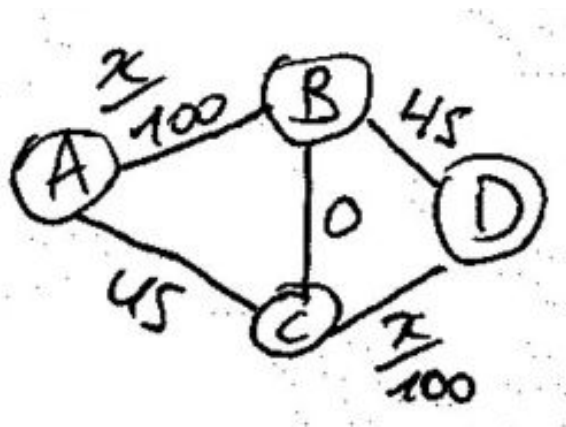
Initial situation:



We have here a **Nash equilibrium**. Both routes are used. If the number of players is 4000, then the cost caused by the equilibrium is:

$$\frac{2000}{100} + 45 = 65 \quad \text{for both routes.}$$

Let's add a B-C road:



Here, we observe that the **Nash equilibrium** is no longer the same as before. Now, all players will take route A-B-C-D, which has a cost of:

$$\frac{4000}{100} + 0 + \frac{4000}{100} = 80.$$

Without **government regulation**, no other route performs better.

5.3 Q3 : Principles of indifference

Principle of Indifference In a **mixed-strategy Nash equilibrium**, each player must be **indifferent** between the pure strategies they randomize over. This means that **each of these strategies** gives them the same expected payoff.

Mixed Strategy A random choice between two pure strategies.

5.4 Q3.2 : Mixed-Strategy equilibrium for penalty kick game

P = dive left:

$$\begin{aligned} 0.58p + 0.95(1 - p) &= 0.93p + 0.70(1 - p) \\ \Rightarrow 0.60p &= 0.25 \quad \Rightarrow p = \frac{10}{24} \end{aligned}$$

Q = kick left:

$$\begin{aligned} -0.58q - 0.93(1 - q) &= -0.95q - 0.70(1 - q) \\ \Rightarrow \frac{1}{2}q &= 0.23 \quad \Rightarrow q = 0.46 \end{aligned}$$

Mixed strategy:

$$\left(\frac{10}{24}; 0.46 \right)$$

5.5 Q4 : Adjacency matrix and random walk matrix

The **adjacency matrix** defines a graph $G = (V, E)$. $A[i][j]$ is equal to 1 if there is an edge from node i to node j ; otherwise, it is 0.

A^2 and A^3 allow us to obtain a matrix that indicates how many paths of length n exist from a node i to a node j . n is the exponent, and $A^n[i][j]$ gives the number of paths of length n from i to j .

The **random walk matrix** (or **transition matrix**) M is defined as:

$$M = DA$$

where A is the **adjacency matrix** and D is a **diagonal matrix** in which each value is $\frac{1}{m}$, with m being the number of edges from each node.

5.6 Q5 :PageRank algorithm in matrix form

Google **PageRank** can be seen as the **principal eigenvector** of the **transition matrix** P of the web graph. It satisfies the equation:

$$\pi \cdot M = \pi$$

Where π is the **PageRank vector** and M the **transition matrix**.

Problems:

- **Dead ends** : Pages with no outgoing links $\Rightarrow$ they absorb all the PageRank score $\Rightarrow$ the score "leaks" and disappears.
- **Spider traps** : Group of pages that only link to each other $\Rightarrow$ the random walker gets stuck $\Rightarrow$ the PageRank score is no longer redistributed.

Teleportation fix : to solve this, random jumps are introduced with a probability of $1 - s$. A scaling is then applied at each iteration: For all pages p :

$$\text{pr}(p) := s \cdot \text{pr}(p) + \frac{1 - s}{n}$$

where n is the **total number of pages**.

Modified PageRank equation: the algorithm becomes a combination of propagated and redistributed scores:

$$\text{pr}(p) = s \cdot (\text{sum of incoming pr scores}) + \frac{1 - s}{n}$$

This process is repeated until the scores converge to a stable state.

5.7 Q6 : Link farm definition (Look question ...)

5.8 Q6.2 : Operation of a link farm

5.9 Q7 : Direct benefit effects

The **direct benefit effect** describes how **the more a product is used by the population, the more valuable it becomes**.

This **added value**, which depends on **the number of users**, is **not intrinsic** to the object.

The telephone: the more people use it, the more valuable it is, because **I can call more people**.

5.10 Q7.2 Tipping points (Look question ...)